Dr. Joshua D. Lothringer

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Research Interests

Observing, modeling, and retrieving planetary, exoplanetary, brown dwarf, and stellar atmospheres.

Positions

Assistant Astronomer 01/2024 - Present
Space Telescope Science Institute
Baltimore, MD

Assistant Research Scientist 01/2025 - Present
William H. Miller III Department of Physics and Astronomy
Johns Hopkins University
Baltimore, MD

Assistant Professor 08/2021-12/2023
Department of Physics
Utah Valley University
Orem, UT

Postdoctoral Fellow 08/2019-08/2021
Department of Physics and Astronomy
Johns Hopkins University
Baltimore, MD

Graduate Assistant/Associate 08/2014-08/2019
Lunar and Planetary Laboratory
University of Arizona
Tucson, AZ

Command Controller 05/2012-08/2014
Mission Operations and Data Systems,
Laboratory for Atmospheric and Space Physics
University of Colorado at Boulder
Boulder, CO

Education

Doctor of Philosophy, Planetary Science 08/2014-08/2019
Master of Science, Planetary Science 12/2016
University of Arizona, Tucson, AZ
Advisor: Prof. Travis Barman
Dissertation: *Characterizing the Atmospheres of Planet Populations: From Sub-Jovian to Ultra-hot Jupiter Exoplanets*

Bachelor of Arts, Astronomy 08/2010-12/2013
University of Colorado, Boulder CO
Concentration: Astrophysics
Minor: Philosophy

Publications
incl. submitted

1. **Lothringer, J. D.**, Lowson, N., Fu, G. "The Library of Exoplanet Atmospheric Composition Measurements: Population Level Trends in Exoplanet Composition with ExoComp", 2026, AJ, 171, 31.
2. **Lothringer, J. D.**, Bennett, K. A., Sing, D. K., et al., "Refractory and Volatile Species in the UV-to-IR Transmission Spectrum of Ultra-hot Jupiter WASP-178b with HST and JWST", 2025, AJ, 169, 274.
3. **Lothringer, J. D.**, Zhou, Y., Apai, D., et al., "Atmospheric Retrievals of the Phase-resolved Spectra of Irradiated Brown Dwarfs WD-0137B and EPIC-2122B", 2024, ApJ, 968, 126.
4. **Lothringer, J.D.**, Sing, D. K., Rustamkulov, Z., et al. "UV absorption by silicate cloud precursors in ultra-hot Jupiter WASP-178b", 2022, *Nature*, 604, 49.
5. **Lothringer, J.D.**, Rustamkulov, Z., Sing, D. K., et al. "A New Window into Planet Formation and Migration: Refractory-to-Volatile Elemental Ratios in Ultra-hot Jupiters", 2021, ApJ, 914, 12.
6. **Lothringer, J.D.**, & Casewell, S. L. "Atmosphere Models of Brown Dwarfs Irradiated by White Dwarfs: Analogs for Hot and Ultrahot Jupiters", 2020, ApJ, 905, 163.
7. **Lothringer, J.D.**, Fu, G., Sing, D. K., & Barman, T. S. "UV Exoplanet Transmission Spectral Features as Probes of Metals and Rainout", 2020, ApJL, 898, L14.
8. **Lothringer, J.D.**, & Barman, T. S. "The PHOENIX Exoplanet Retrieval Algorithm and Using H^- Opacity as a Probe in Ultrahot Jupiters", 2020, AJ, 159, 289.
9. **Lothringer, J.D.**, & Barman, T. "The Influence of Host Star Spectral Type on Ultra-hot Jupiter Atmospheres", 2019, ApJ, 876, 69.
10. **Lothringer, J.D.**, Barman, T., & Koskinen, T. "Extremely Irradiated Hot Jupiters: Non-oxide Inversions, H^- Opacity, and Thermal Dissociation of Molecules", 2018, ApJ, 866, 27.
11. **Lothringer, J.D.**, Benneke, B., Crossfield, I. J. M., et al. "An HST/STIS Optical Transmission Spectrum of Warm Neptune GJ 436b", 2018, AJ, 155, 66.
12. Wang, L.-C., Rustamkulov, Z., Sing, D. K., et al., "A Comprehensive Analysis of the Panchromatic Transmission Spectrum of the Hot-Saturn WASP-96 b: Nondetection of Haze, Possible Sodium Limb Asymmetry, Stellar Characterization, and Formation History", 2026, AJ, 171, 147.
13. Baldwin, A., **Lothringer, J. D.**, Dos Santos, L. A., et al., "A High-Resolution NUV Transmission Spectrum of KELT-9b: Mg II and Fe II

Escaping from the Hottest Known Giant Planet", 2025, arXiv e-prints, arXiv:2512.19662.

14. Crossfield, I. J. M., Ahrer, E.-M., Brande, J., et al., "Mapping the SO_2 Shoreline in Gas Giant Exoplanets", 2025, ApJ, 994, 184.
15. Ashtari, R., Collins, S., Splinter, J., et al., "Heat Reveals What Clouds Conceal: Global Carbon & Longitudinally Asymmetric Chemistry on LTT 9779 b", 2025, arXiv e-prints, arXiv:2510.04863.
16. Chachan, Y., **Lothringer, J.D.**, Inglis, J., et al., "Strong NUV Refractory Absorption and Dissociated Water in the Hubble Transmission Spectrum of the Ultra Hot Jupiter KELT-20 b", 2025, AJ, 170, 234.
17. Bennett, K. A., MacDonald, R. J., Peacock, S., et al., "Additional JWST/NIRSpec Transits of the Rocky M Dwarf Exoplanet GJ 1132 b Reveal a Featureless Spectrum", 2025, AJ, 170, 205.
18. Fu, G., Mukherjee, S., Stevenson, K. B., et al., "Overcast Mornings and Clear Evenings in Hot Jupiter Exoplanet Atmospheres", 2025, ApJL, 989, L17.
19. Evans-Soma, T. M., Sing, D. K., Barstow, J. K., et al., "SiO and a super-stellar C/O ratio in the atmosphere of the giant exoplanet WASP-121 b", 2025, Nature Astronomy, 9, 845.
20. Fu, G., Stevenson, K. B., Sing, D. K., et al., "Statistical Trends in JWST Transiting Exoplanet Atmospheres", 2025, ApJ, 986, 1.
21. Gapp, C., Evans-Soma, T. M., Barstow, J. K., et al., "WASP-121 b's Transmission Spectrum Observed with JWST/NIRSpec G395H Reveals Thermal Dissociation and SiO in the Atmosphere", 2025, AJ, 169, 341.
22. Balmer, W. O., Kammerer, J., Pueyo, L., et al., "JWST-TST High Contrast: Living on the Wedge, or, NIRCam Bar Coronagraphy Reveals CO_2 in the HR 8799 and 51 Eri Exoplanets' Atmospheres", 2025, AJ, 169, 209.
23. Amaro, R. C., Apai, D., Zhou, Y., et al., "Phase-resolved Hubble Space Telescope WFC3 Spectroscopy of the Weakly Irradiated Brown Dwarf GD 1400 and Energy Redistribution–Irradiation Trends in Six White Dwarf–Brown Dwarf Binaries", 2025, ApJ, 979, 231.
24. Bennett, K. A., Sing, D. K., Stevenson, K. B., et al., "An HST Transmission Spectrum of the Closest M Dwarf Transiting Rocky Planet LTT 1445Ab", 2025, AJ, 169, 111.
25. Pelletier, S., Benneke, B., Chachan, Y., et al., "CRIRES$^+$ and ESPRESSO Reveal an Atmosphere Enriched in Volatiles Relative to Refractories on the Ultrahot Jupiter WASP-121b", 2025, AJ, 169, 10.
26. Sing, D. K., Evans-Soma, T. M., Rustamkulov, Z., et al., "An Absolute Mass, Precise Age, and Hints of Planetary Winds for WASP-121A and b from a JWST NIRSpec Phase Curve", 2024, AJ, 168, 231.
27. Iskandarli, L., Farihi, J., **Lothringer, J. D.**, et al., "Novel constraints on companions to the Helix nebula central star", 2024, MNRAS, 534, 3498.

28. French, J. R., Casewell, S. L., Amaro, R. C., et al., "The only inflated brown dwarf in an eclipsing white dwarf-brown dwarf binary: WD1032+011B", 2024, MNRAS, 534, 2244.

29. Fu, G., Welbanks, L., Deming, D., et al., "Hydrogen sulfide and metal-enriched atmosphere for a Jupiter-mass exoplanet", 2024, *Nature*, 632, 752.

30. Carter, A. L., May, E. M., Espinoza, N., et al., "A benchmark JWST near-infrared spectrum for the exoplanet WASP-39 b", 2024, Nature Astronomy, 8, 1008.

31. Bell, T. J., Crouzet, N., Cubillos, P. E., et al., "Nightside clouds and disequilibrium chemistry on the hot Jupiter WASP-43b", 2024, Nature Astronomy, 8, 879.

32. Amaro, R. C., Apai, D., Lew, B. W. P., et al., "Time-resolved Hubble Space Telescope Wide Field Camera 3 Spectrophotometry Reveals Inefficient Day-to-night Heat Redistribution in the Highly Irradiated Brown Dwarf SDSS 1557B", 2024, ApJ, 966, 4.

33. Kirk, J., Stevenson, K. B., Fu, G., et al., "JWST/NIRCam Transmission Spectroscopy of the Nearby Sub-Earth GJ 341b", 2024, AJ, 167, 90.

34. Powell, D., Feinstein, A. D., Lee, E. K. H., et al., "Sulfur dioxide in the mid-infrared transmission spectrum of WASP-39b", 2024, *Nature*, 626, 979.

35. Brande, J., Crossfield, I. J. M., Kreidberg, L., et al., "Clouds and Clarity: Revisiting Atmospheric Feature Trends in Neptune-size Exoplanets", 2024, ApJL, 961, L23.

36. Reggiani, H., Galarza, J. Y., Schlaufman, K. C., et al., "Insight into the Formation of β Pic b through the Composition of Its Parent Protoplanetary Disk as Revealed by the β Pic Moving Group Member HD 181327", 2024, AJ, 167, 45.

37. May, E. M., MacDonald, R. J., Bennett, K. A., et al., "Double Trouble: Two Transits of the Super-Earth GJ 1132 b Observed with JWST NIRSpec G395H", 2023, ApJL, 959, L9.

38. Lustig-Yaeger, J., Fu, G., May, E. M., et al., "A JWST transmission spectrum of the nearby Earth-sized exoplanet LHS 475 b", 2023, Nature Astronomy, 7, 1317.

39. Roy, P.-A., Benneke, B., Piaulet, C., et al., "Water Absorption in the Transmission Spectrum of the Water World Candidate GJ 9827 d", 2023, ApJL, 954, L52.

40. Coria, D. R., Crossfield, I. J. M., **Lothringer, J.D.**, et al. "The Missing Link: Testing Galactic Chemical Evolution Models with the First Multi-isotopic Abundances in Solar Twin Stars", 2023, ApJ, 954, 121.

41. Coulombe, L.-P., Benneke, B., Challener, R., et al. "A broadband thermal emission spectrum of the ultra-hot Jupiter WASP-18b", 2023, *Nature*, 620, 292.

42. van Sluijs, L., Birkby, J. L., **Lothringer, J.D.**, et al., "Carbon monoxide emission lines reveal an inverted atmosphere in the ultra hot Jupiter WASP-33 b consistent with an eastward hot spot", 2023, MNRAS, 522, 2145.

43. Tsai, S.-M., Lee, E. K. H., Powell, D., et al., "Photochemically produced SO_2 in the atmosphere of WASP-39b", 2023, *Nature*, 617, 483.

44. Grant, D., **Lothringer, J.D.**, Wakeford, H. R., et al., "Detection of Carbon Monoxide's 4.6 Micron Fundamental Band Structure in WASP-39b's Atmosphere with JWST NIRSpec G395H", 2023, ApJL, 949, L15.

45. Moran, S. E., Stevenson, K. B., Sing, D. K., et al., "High Tide or Riptide on the Cosmic Shoreline? A Water-rich Atmosphere or Stellar Contamination for the Warm Super-Earth GJ 486b from JWST Observations", 2023, ApJL, 948, L11.

46. Amaro, R. C., Apai, D., Zhou, Y., et al., "Hotter than Expected: Hubble Space Telescope (HST)/WFC3 Phase-resolved Spectroscopy of a Rare Irradiated Brown Dwarf with Strong Internal Heat Flux", 2023, ApJ, 948, 129.

47. Gressier, A., Lecavelier des Etangs, A., Sing, D. K., et al., "The Hubble PanCET program: The near-ultraviolet transmission spectrum of WASP-79b", 2023, A&A, 672, A34.

48. Feinstein, A. D., Radica, M., Welbanks, L., (**Lothringer, J.D. 1st tier**) et al., "Early Release Science of the exoplanet WASP-39b with JWST NIRISS", 2023, *Nature*, 614, 670.

49. Alderson, L., Wakeford, H. R., Alam, M. K., (**Lothringer, J.D. 1st tier**) et al., "Early Release Science of the exoplanet WASP-39b with JWST NIRSpec G395H", 2023, *Nature*, 614, 664.

50. Rustamkulov, Z., Sing, D. K., Mukherjee, S., (**Lothringer, J.D. 1st tier**) et al., "Early Release Science of the exoplanet WASP-39b with JWST NIRSpec PRISM", 2023, *Nature*, 614, 659.

51. Ahrer, E.-M., Stevenson, K. B., Mansfield, M., et al., "Early Release Science of the exoplanet WASP-39b with JWST NIRCam", 2023, *Nature*, 614, 653.

52. JWST Transiting Exoplanet Community Early Release Science Team, Ahrer, E.-M., Alderson, L., (**Lothringer, J.D. 1st tier**) et al., "Identification of carbon dioxide in an exoplanet atmosphere", 2023, *Nature*, 614, 649.

53. Mikal-Evans, T., Sing, D. K., Dong, J., et al. "A JWST NIRSpec Phase Curve for WASP-121b: Dayside Emission Strongest Eastward of the Substellar Point and Nightside Conditions Conducive to Cloud Formation", 2023, ApJL, 943, L17.

54. Chachan, Y., Knutson, H. A., **Lothringer, J.D.**, & Blake, G. A. "Breaking Degeneracies in Formation Histories by Measuring Refractory Content in Gas Giants", 2023, ApJ, 943, 112.

55. Kasper, D., Bean, J. L., Line, M. R., et al., "Unifying High- and Low-resolution Observations to Constrain the Dayside Atmosphere of KELT-20b/MASCARA-2b", 2023, AJ, 165, 7.

56. Fu, G., Espinoza, N., Sing, D. K., et al., "Water and an Escaping Helium Tail Detected in the Hazy and Methane-depleted Atmosphere of HAT-P-18b from JWST NIRISS/SOSS", 2022, ApJL, 940, L35.

57. Brande, J., Crossfield, I. J. M., Kreidberg, L., et al., "A Mirage or an Oasis? Water Vapor in the Atmosphere of the Warm Neptune TOI-674 b", 2022, AJ, 164, 197.

58. Kreidberg, L., Mollière, P., Crossfield, I. J. M., et al., "Tentative Evidence for Water Vapor in the Atmosphere of the Neptune-sized Exoplanet HD 106315c", 2022, AJ, 164, 124.

59. Buzard, C., Casewell, S. L.,**Lothringer, J.D.**, & Blake, G. A., "Near-infrared Spectra of the Inflated Post-common Envelope Brown Dwarf NLTT 5306 B", 2022, AJ, 163, 262.

60. Gibson, N. P., Nugroho, S. K., **Lothringer, J.D.**, Maguire, C., & Sing, D. K., "Relative abundance constraints from high-resolution optical transmission spectroscopy of WASP-121b, and a fast model-filtering technique for accelerating retrievals", 2022, MNRAS, 512, 4618.

61. Fu, G., Sing, D. K., Deming, D., et al., "The Hubble PanCET Program: Emission Spectrum of Hot Jupiter HAT-P-41b", 2022, AJ, 163, 190.

62. Reggiani, H., Schlaufman, K. C., Healy, B. F., **Lothringer, J.D.**, & Sing, D. K., "Evidence that the Hot Jupiter WASP-77 A b Formed Beyond Its Parent Protoplanetary Disk's H2O Ice Line", 2022, AJ, 163, 159.

63. Bruno, G., Lewis, N. K., Valenti, J. A., et al., "Hiding in plain sight: observing planet-starspot crossings with the James Webb Space Telescope", 2022, MNRAS, 509, 5030.

64. Fu, G., Sing, D. K., **Lothringer, J.D.**, et al., "Strong H_2O and CO Emission Features in the Spectrum of KELT-20b Driven by Stellar UV Irradiation", 2022, ApJL, 925, L3.

65. Zhou, Y., Apai, D., Tan, X., et al., "HST/WFC3 Complete Phase-resolved Spectroscopy of White-dwarf-brown-dwarf Binaries WD 0137 and EPIC 2122", 2022, AJ, 163, 17.

66. Fu, G., Deming, D., May, E., et al., "The Hubble PanCET program: Transit and Eclipse Spectroscopy of the Hot-Jupiter WASP-74b", 2021, AJ, 162, 271.

67. Sainsbury-Martinez, F., Casewell, S. L., **Lothringer, J.D.**, Phillips, M. W., & Tremblin, P., "Exploring deep and hot adiabats as a potential solution to the radius inflation problem in brown dwarfs. Long-timescale models of the deep atmospheres of KELT-1b, Kepler-13Ab, and SDSS1411B", 2021, A&A, 656, A128.

68. Merritt, S. R., Gibson, N. P., Nugroho, S. K., et al., "An inventory of atomic species in the atmosphere of WASP-121b using UVES high-resolution spectroscopy", 2021, MNRAS, 506, 3853.

69. Fu, G., Deming, D., **Lothringer, J.D.**, et al., "The Hubble PanCET Program: Transit and Eclipse Spectroscopy of the Strongly Irradiated Giant Exoplanet WASP-76b", 2021, AJ, 162, 108.

70. Wilson, J., Gibson, N. P., **Lothringer, J.D.**, et al., "Gemini/GMOS optical transmission spectroscopy of WASP-121b: signs of variability in an ultra-hot Jupiter?", 2021, MNRAS, 503, 4787.

71. Mikal-Evans, T., Crossfield, I. J. M., Benneke, B., et al., "Transmission Spectroscopy for the Warm Sub-Neptune HD 3167c: Evidence for Molecular Absorption and a Possible High-metallicity Atmosphere", 2021, AJ, 161, 18.

72. Guo, X., Crossfield, I. J. M., Dragomir, D., et al., "Updated Parameters and a New Transmission Spectrum of HD 97658b", 2020, AJ, 159, 239.

73. Gibson, N. P., Merritt, S., Nugroho, S. K., et al., "Detection of Fe I in the atmosphere of the ultra-hot Jupiter WASP-121b, and a new likelihood-based approach for Doppler-resolved spectroscopy", 2020, MNRAS, 493, 2215.

74. Turner, J. D., de Mooij, E. J. W., Jayawardhana, R., et al., "Detection of Ionized Calcium in the Atmosphere of the Ultra-hot Jupiter KELT-9b", 2020, ApJL, 888, L13.

75. Benneke, B., Wong, I., Piaulet, C., et al., "Water Vapor and Clouds on the Habitable-zone Sub-Neptune Exoplanet K2-18b", 2019, ApJL, 887, L14.

76. Benneke, B., Knutson, H. A., **Lothringer, J.D.**, et al., "A sub-Neptune exoplanet with a low-metallicity methane-depleted atmosphere and Mie-scattering clouds", 2019, Nature Astronomy, 3, 813.

77. Steinrueck, M. E., Parmentier, V., Showman, A. P., **Lothringer, J.D.**, & Lupu, R. E., "The Effect of 3D Transport-induced Disequilibrium Carbon Chemistry on the Atmospheric Structure, Phase Curves, and Emission Spectra of Hot Jupiter HD 189733b", 2019, ApJ, 880, 14.

78. Crossfield, I. J. M., **Lothringer, J.D.**, Flores, B., et al., "Unusual Isotopic Abundances in a Fully Convective Stellar Binary", 2019, ApJL, 871, L3.

79. Fossati, L., Koskinen, T., **Lothringer, J.D.**, et al., "Extreme-ultraviolet Radiation from A-stars: Implications for Ultra-hot Jupiters", 2018, ApJL, 868, L30.

80. Bean, J. L., Stevenson, K. B., Batalha, N. M., et al., "The Transiting Exoplanet Community Early Release Science Program for JWST", 2018, PASP, 130, 114402.

81. Kilpatrick, B. M., Cubillos, P. E., Stevenson, K. B., et al., "Community Targets of JWST's Early Release Science Program: Evaluation of WASP-63b", 2018, AJ, 156, 103.

82. Bell, T. J., Nikolov, N., Cowan, N. B., et al., "The Very Low Albedo of WASP-12b from Spectral Eclipse Observations with Hubble, 2017, ApJL, 847, L2.

83. Stevenson, K. B., Lewis, N. K., Bean, J. L., et al., "Transiting Exoplanet Studies and Community Targets for JWST's Early Release Science Program", 2016, PASP, 128, 094401.

84. Crossfield, I. J. M., Ciardi, D. R., Petigura, E. A., et al., "197 Candidates and 104 Validated Planets in K2's First Five Fields", 2016, ApJS, 226, 7.

Proceedings and Other Publications

1. Ardila, D. R., et al. "The UV-SCOPE Mission: Ultraviolet Spectroscopic Characterization Of Planets and their Environments", Proceedings of the SPIE. id. 1218104. arXiv:2208.09547.
2. **Lothringer, J. D.** "Stellar specific intensity models used in 'Hiding in plain sight: observing planet-starspot crossings with the James Webb Space Telescope'", 2021, Zenodo Software package, id. 5609421.
3. **Lothringer, J. D.** et al. "Uncertainties in Low-Count STIS Spectra", 2025, HST STIS Instrument Science Report 2025-04. arXiv:2601.18910.
4. Alam, M., et al. "HST Notebooks First Release", Zenodo. Version v1.0.0. 10.5281/zenodo.17179608

Invited Talks and Seminars

1. Exocoffee Seminar on Lothringer et al. 2025, Max Planck Institute for Astronomy, Heidelberg, Germany. May. 2025.
2. "Ultra-hot Jupiters as Unique Windows into Formation though the UV-IR Transit Spectrum of WASP-178b", The Great Link Workshop, Max Planck Institute for Astronomy, Heidelberg, Germany. Jul. 2024.
3. "Talk with Tuminez", University-wide Town Hall with President Astrid Tuminez, Utah Valley University, Orem, UT. Sep. 2023.
4. Convocation Lecture Series & Science Division Seminar, Snow College, Ephriam, UT. Aug. 2023.
5. Astronomy Colloquium, Anton Pannekoek Institute for Astronomy, University of Amsterdam. Nov. 2022.
6. Physics and Astronomy Colloquium, Brigham Young University, Provo, UT. Oct. 2022.
7. The Brown Dwarf to Exoplanet Connection in the Era of JWST Splinter Session at Exoplanets IV. Las Vegas. May 2022.
8. Physics Colloquium. Department of Physics, Utah Valley University, Orem, UT. Sep. 2021.
9. HotSci. Space Telescope Science Institute. Virtual. Jul. 2021.
10. Exoplanet Lunch. Center for Astrophysics, Harvard University. Virtual. Jan. 2021.
11. Exoplanet Journal Club. Jet Propulsion Laboratory. Virtual. Jan. 2021.
12. Star and Planet Seminar. Imperial College London. Virtual. Oct. 2020.
13. Exocoffee. Max Planck Institute for Astronomy. Virtual. May. 2020.
14. Exoplanet Tea. Massachusetts Institute of Technology. Cambridge, MA. Oct. 2019.
15. Exoplanet Lunch. Center for Astrophysics, Harvard University. Cambridge, MA. Oct. 2019.

16. Wine & Cheese Seminar. Center for Astrophysical Sciences, Johns Hopkins University, Baltimore, MD. Sep. 2019.
17. Theoretical Astrophysics Program Graduate Research Prize Talk. University of Arizona. Tucson, AZ. Apr. 2019.
18. Exoplanet Seminar. DTU Space. Lyngby, Denmark. Feb. 2019.
19. Star and Planet Formation Seminar. Max Planck Institute for Astronomy. Heidelberg, Germany. Jul. 2016.

Select Contributed Conference Presentations

1. "The UVOIR Transit Spectrum of Ultra-hot Jupiter WASP-178b with HST and JWST." 245th AAS Winter Meeting. National Harbor, MD. Jan. 2025. Oral Presentation.
2. "Atmospheric Retrievals of the Phase-resolved Spectra of Two Hot and Ultra-hot Jupiter Analogs." 241st AAS Winter Meeting. Seattle, WA. Jan. 2023. Oral Presentation.
3. "The UV Transmission Spectrum of Ultra-hot Jupiter WASP-178b." Exoplanets IV. Las Vegas. May 2022. Oral Presentation.
4. "The Importance of UV Opacity in Extremely Irradiated Objects." Stars and Planets in the UV. Virtual. May. 2021. Oral Presentation.
5. "Re-Interpreting UV-Optical Transmission Spectra of Hot and Ultra-Hot Jupiters." 237th AAS Winter Meeting. Virtual. Jan. 2021. Oral Presentation.
6. "Understanding Ultra-hot Jupiters Through Irradiated Brown Dwarfs." 235th AAS Winter Meeting. Honolulu, HI. Jan. 2020. Oral Presentation.
7. "Highly Irradiated Brown Dwarfs as High-mass Ultra-hot Jupiters." BDExoCon. University of Delaware. Newark, DE. Oct. 2019. Oral Presentation.
8. "Characterizing the Atmospheres of Exoplanet Populations: From Sub-Jovian to Ultra-hot Jupiter Exoplanets." American Astronomical Society Winter Meeting. Seattle, WA. Jan. 2019. Oral Presentation.
9. "Modeling the Most Extreme Jovian Atmospheres." Exoplanets Around Hot Stars. Vanderbilt University, Nashville, TN. Jun. 2018. Oral Presentation.
10. "Self-Consistent Atmosphere Models of the Most Extreme Hot Jupiters." American Astronomical Society Winter Meeting. Washington D.C. Jan. 2018. Oral Presentation.

Honors, Awards, and Grants
+$2,300,000

- **PI, NSF S-STEM Program** *2022–2023*
  "Pro-STEM: Promoting Engagement in Chemistry, Physics, and Earth Sciences"
  A six-year, $1.5M scholarship, mentorship, and research program, awarding scholarships to 25+ students in Chemistry, Physics, and Earth Sciences.

- **PI/Co-PI, *James Webb Space Telescope* Programs**
  - Program 2055 ($100,211) *9.1 hours*
    "Tracing Hot Jupiter Formation and Migration with Volatile and Refractory Element Ratios"
  - Program 2288 ($113,998) *7.4 hours*
    Co-PI: Jeff Valenti
    "Formation and Impact of Silicate Clouds on L Dwarfs"
  - Program 7255 ($308,468) *52.3 hours*
    PI: Kevin Stevenson
    "A Day In The Life of KELT-9b"
- **PI, *Hubble Space Telescope* Programs**
  - Program 16086 ($86,995) *10 orbits*
    "Comparing Escaping Metals and Heat Deposition in Ultra-hot Jupiters"
  - Program 16142 ($99,319) *AR Theory*
    "The First Grid of White-Dwarf-Irradiated Brown Dwarf Atmosphere Models"
  - Program 16270 ($63,530) *20 orbits*
    "Heavy Metal Bands: A Study of Escaping Ions from the Hottest Jovian Atmospheres"
  - Program 16450 ($45,465) *10 orbits*
    "Measuring the Rock-to-Ice Ratio in an Exoplanet"

PI of 3 Calibration *Hubble Space Telescope* Programs
Co-I on 11 *James Webb Space Telescope* Programs *500+ hours*
Co-I on 21 *Hubble Space Telescope* Programs *300+ orbits*
Co-I on 1 *Spitzer Space Telescope* Program *61 hours*
Co-I on 1 NASA XRP Program (PI: Schlaufman) 2022-2025
Director's Discretionary Research Funds: CHEXO Funding ($3,400) 2024
College of Science Dean's Award of Excellence for Scholarship ($2,750) 2023
Scholarly Activities Committee Dissemination Grant ($1,344) 2022
Theoretical Astrophysics Program Graduate Research Prize ($500) 2019
Galileo Circle Scholar ($2,000) 2019
Galileo Circle Scholar ($1,000) 2016
1st Place - The Art of Planetary Science - Data Art Category 2015
Graduate and Professional Student Council Travel Grant ($250) 2015
2015 Sagan Workshop Travel Grant ($700) 2015
Science Phoenix Award - SORCE Mission Operations 2014

Technical Experience

HST/STIS Instrument Team 2024-Present
- Pipeline Block Lead
- Exposure Time Calculator Working Group
- Contact Scientist
- PI/co-I for 5 calibration programs

	JWST Science Policy Division	2024-Present
	JWST+HST Rocky Worlds DDT Core Implementation Team	2025-Present
Observing Experience	James Webb Space Telescope - NIRSpec, NIRCam, and MIRI	300+ hours
	Hubble Space Telescope - STIS and WFC3	300+ orbits
	MMT - SWIRC and ARIES	12 nights
	Sommers-Bausch Observatory (CU-Boulder) - Optical CCD	9 nights
	W.M. Keck Observatory - OSIRIS	2 nights
	Large Binocular Telescope - LMIRCam	1 night
	Morris W. Offit Telescope (JHU) - Optical CCD	1 night
Mentorship	Post-Baccalaureate Mentor:	
	• **Seti Norris** - STScI SASP+Intern	2024-Present
	-Co-Author on Lothringer et al. 2024	
	Undergraduate Research Mentor:	
	• **Ja'Heim Goodwin** - STScI SASP+NAC	2024-Present
	• **Brian Kehoe-Seamons** - Utah Valley University	2022-Present
	-Utah Space Grant Recipient	
	-Co-Author on Lothringer et al. 2024	
	• **Austin Baldwin** - Utah Valley University	2021-Present
	-UVU URSCA Grant Recipient	
	-Utah Space Grant Recipient	
	• **Audrey Elison** - Utah Valley University	2023
	• **Brayden Roberts** - Utah Valley University	2022-2024
	-Utah Space Grant Recipient	
	• **Autumn Winch** - Bryn Mawr College	2020-2022
	-Senior Thesis *-Co-Author on Lothringer et al. 2022*	
	-Clinical Research Coordinator, University of Pennsylvania	
	Dissertation Committee Member	
	• Rachael Amaro (University of Arizona)	08/2024
	• Jegug Ih (University of Maryland)	06/2024
Teaching (UVU)	Instructor - ASTR-4100 - Exoplanets and Brown Dwarfs	Fall 2023
	Instructor - ASTR-1080 - Life in the Universe (3 sections)	2022-2023
	Instructor - PHSC-1000 - Survey of Physical Science	Spring 2022
	Instructor - PHYS-489R - Undergraduate Research (6 sections)	2022-2023
	Instructor - ASTR-1040 - Elementary Astronomy (12 Sections)	2021-2023
Other Teaching Experience	Guest Instructor - Exoplanetary Astrophysics (UMD)	2024
	Guest Instructor - Planets, Life, and the Universe (JHU)	2020
	Co-Instructor - Exoplanets & Their Atmospheres (JHU)	2020
	JHU Teaching Academy Certificate	2020
	JHU Summer Teaching Institute Workshop	2020
	LPL Incoming Graduate Student Mentor	2017-2019
	Pima Community College GED Prep Math Tutor	2015-2016
	Graduate Teaching Assistant and Guest Lecturer — PTYS 170B2	Fall 2014

Service and Other Experience

AAS Journals Reviewer
Astronomy & Astrophysics Reviewer
Nature Astronomy Reviewer
Nature Reviewer
Canada Innovation Fund Expert Reviewer
Hubble Space Telescope Proposal Reviewer
Canadian Time Allocation Committee Reviewer
NSF Review Panelist
NASA Review Panel Executive Secretary

Chesapeake Bay Area Exoplanet Meeting Co-Organizer	2024–Present
STARGATE Collaboration Executive Committee	2023–Present
STScI Science Evaluation Committee	2025
STScI Atmospheric Escape Workshop Co-Organizer	2025
Aspen CfP JWST Thermal Emission SOC	Spring 2025
STScI-JHU ExoJamboree SOC/LOC	Fall 2024
STScI Summer Program Mentor	2024
UVU Physics Department Advisory Board	2023-2024
UV-SCOPE MIDEX Mission Concept Science Team	2020-2023
JWST ERS Working Group	2017-2024
JHU/STScI Undergraduate Summer Program Organizer	2020, 2021
AAS Chambliss Poster Award Judge	2019, 2020
LPL Men's Diversity and Inclusion Auxiliary	2016-2019
LPL Conference Organizing Committee	2015-2017
Visiting Student - Max Planck Institute for Astronomy	06-07/2016
Graduate and Professional Student Council Travel Grant Judge	2015

Select Press Coverage & Interviews

The (Unofficial) Exoplanet Dashboard
- Space.com

JWST Transiting Exoplanet Community Early Release Science
- UVU Press Release
- Fox 31 News Good Day Utah at 6AM
- KUER 90.1, Utah NPR

Lothringer & Sing et al. 2022
- NASA/Hubble Press Release
- Sky & Telescope
- The Bad Astronomer, SyFy Wire
- The Miami Herald

Benneke et al. 2019
- SkyMania
- IndiaTV News

In the Community
- UVU Wins $1.5M NSF S-STEM Scholarship Program
- Hutchings Museum Institute JWST First Images Event
- Hutchings Museum Institute JWST Launch Celebration

- MAVEN Mission to Mars Launch
- No Man's Sky Video Game

Outreach

JWST Subject Matter Expert	2021-Present
- Salt Lake County Library Lecture Series (x2)	
- UVU Colloquium (x2)	
- Clarke Planetarium JWST Program	
- Utah Valley Astronomy Club (x2)	
- Hutchings Museum Institute First Image Event	
- Hutchings Museum Institute Launch Celebration	
Latinos of Tomorrow Panelist	2023
Guest Consultant - Ranger Rick Magazine	2022
STScI Outreach Program	2019-2021
-Space Astronomy Summer Program Presenter and Organizer	
-Easy as Pi - Society of American Military Engineers	
Reddit /r/AskScience Panel Member	2015-2018
LPL Outreach Program	2014-2019
-Summer Science Saturdays	
-Tucson Festival of Books	
-Art of Planetary Science	
-Bennuval: An Evening of Space, Art, and Music	
Exoplanet Lecture Series, Flandrau Planetarium, Tucson, AZ	Apr. 2018
Tucson Amateur Astronomy Association	Mar. 2018
The American International School of Muscat Science Expert	2018, 2019
Chaparral High School Career Expert	2018
The American International School of Muscat, Muscat, Oman	Jan. 2014
LASP MAVEN Launch Outreach	2014

Professional Affiliations

International Astronomical Union	Since 2025
American Astronomical Society	Since 2014
Phi Beta Kappa Member	Since 2014
Planetary Society Member	Since 2011

Skills

IDL, Python, Fortran, Perl, Bash, Matlab, and Mathematica